Aditya Makkar

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Education

New York University 2024 – present
Ph.D. (reinforcement learning and multi-agent systems)
Advisor: Eugene Vinitsky

Columbia University 2019 – 2022
M.S. and M.Phil. in Electrical Engineering
(Bayesian machine learning and nonparametric statistics; thesis link)

Indian Institute of Technology Delhi 2012 – 2016
B.Tech. in Mechanical Engineering
(thesis on operations research and combinatorial optimization)

Relevant Coursework (for all of the above): Abstract Analysis, Abstract Algebra, Topology, Probability Theory, Complex Analysis, Linear Algebra, Theoretical Statistics, High-dimensional Statistics, Empirical Process Theory, Concentration of Measure, Analysis of Algorithms, Randomized Algorithms, Robotics, Control Theory, Multi-agent Learning.

Research Interests

- Developing new ways of thinking about systems in which capable agents interact with each other.
- Foundations of reinforcement learning with a focus on challenges like exploration, non-stationary environments, and partial observability.
- I try to read broadly and incorporate that thinking into my research. In particular, I heavily employ ideas from game theory, dynamical systems, statistical mechanics, stochastic control, and statistics.

Work Experience

Amazon, Supply Chain Optimization (SCOT) (Applied Science Intern) 2026
Applied reinforcement learning (+ convex optimization) to improve forecasting accuracy during high-volume events like Prime Day and holidays.

Morgan Stanley (Quant Associate) 2023
Developed algorithms to price and trade interest rate products (US Treasury bonds, swaps, futures and options).

Intercontinental Exchange (Data Scientist) 2023
Priced New York Stock Exchange-listed stocks/ETFs and less liquid fixed income products.

Goldman Sachs (Quantitative Strategist) 2016 – 2019
Researched and developed ML models to process firm-wide data and flag anomalous activities; worked with Howard Karloff.

Peer-Reviewed Research Papers

- **Active Learning for Constrained Linear Regression**
  Aarshvi Gajjar, Syamantak Kumar, Aditya Makkar, Christopher Musco, Yihan Zhou; [alphabetic ordering]
  NeurIPS 2026 (spotlight presentation)
  Computationally and statistically efficient algorithms for constrained active regression using randomized leverage score sampling.
- **Putting the Spotlight on the Initial State Distribution**
  Aditya Makkar, Aarshvi Gajjar, and Eugene Vinitsky
  Finding the Frame Workshop, Reinforcement Learning Conference 2025
  Theoretically characterizing the importance of where the agent starts in the RL setting.

Ongoing Research Projects

- **Representation Learning in Mean-field Reinforcement Learning** [arxiv link]
  The mean-field approximation allows tractable analysis of multi-agent systems. We develop a framework for representation learning in mean-field RL when there is an underlying unknown low-dimensional latent structure. Paper submitted to ICLR 2027.
- **A dynamical systems perspective on AI safety**
  Dynamical systems theory and game theory provide a powerful language to model real constantly changing systems of AI agents. This work provides a theoretical framework on how to think about AI safety.
- **Reset-based Exploration in Reinforcement Learning**
  We study reset-based exploration, where the agent can reset to any previously visited state; this can provably reduce sample complexity. We are developing a JAX library implementing a suite of reset-based exploration algorithms that find an optimal policy significantly faster than standard RL algorithms in benchmark environments.

Awards

- **IIT Delhi Semester Merit Award** (twice): Top 7% of class. Fall 2012 and Spring 2013
- **Indian National Mathematical Olympiad** (INMO) finalist (twice): 3rd rank in the state of Uttar Pradesh in the Regional Mathematics Olympiad (RMO). 2012
- **KVPY fellowship**: Awarded to 250 students nationally by the Government of India. 2012
- **NTSE fellowship**: Awarded to 1000 students nationally by the Government of India. 2008

Miscellaneous

- **Teaching Experience:** TA for graduate course in deep learning (Fall 2025, Fall 2024), digital signal processing (Fall 2022), and graduate introduction to machine learning (Spring 2021, Fall 2021).
  Guest lecture on the general theory of stochastic processes in the Ph.D. course for probability theory taught by Professor Ioannis Karatzas (notes here).
- **Programming:** Proficient in JAX and PyTorch.
- **Reviewing:** I have reviewed for NeurIPS (2026, 2025, 2024, 2023, 2022, 2021), RLC (2026), ICML (2025, 2024, 2022, 2020), ICLR (2024, 2023, 2022), AISTATS (2024, 2023, 2022) and AAAI (2021).